

Formula Sheet

SUMMING AMP

INV:

$$V_o = - \left(\frac{R_F \times V_1}{R_1} + \frac{R_F \times V_2}{R_2} + \frac{R_F \times V_3}{R_3} \right)$$

NON-INV:

$$V_o = \left(1 + \frac{R_F}{R_i} \right) \left(V_1 \times \frac{R_2}{R_1 + R_2} + V_2 \times \frac{R_1}{R_1 + R_2} \right)$$

DIFFERENCE AMP

$$e_{out} = \frac{R_f}{R_i} (e_{in+} - e_{in-})$$

INSTRUMENTATION AMP

$$V_o = \left(1 + \frac{2R_1}{R_4} \right) \left(\frac{R_3}{R_2} \right) (V_2 - V_1)$$

Composite Amp

$$I_{out} = \frac{V_{ref}}{10k\Omega} \times \frac{code}{256_{10}}$$

Voltage/Current Parameter Calculations

	Average Value	RMS value
→ DC waveform $V(t) = V_p$	$V_{avg} = V_p$	$V_{rms} = V_p$
→ Sinusoidal $V(t) = V_p \sin \theta$ $V(t) = V_p \sin \omega t$	$V_{avg} = \frac{1}{2\pi} \int_0^{2\pi} V_p \sin \theta d\theta$	$V_{rms} = \frac{V_p}{\sqrt{2}}$
→ half wave rectifier $V(t) = \begin{cases} V_p \sin \theta & 0 \leq \theta < \pi \\ 0 & \pi \leq \theta < 2\pi \end{cases}$	$V_{avg} = \frac{V_p}{\pi}$	$V_{rms} = \frac{V_p}{2}$

	Average value	RMS value
→ Full wave Rectifier		
$V(t) = \begin{cases} V_p \sin \theta & 0 \leq \theta < \pi \\ -V_p \sin \theta & \pi \leq \theta < 2\pi \end{cases}$	$V_{avg} = \frac{2V_p}{\pi}$	$V_{rms} = \frac{V_p}{\sqrt{2}}$
→ Rectangular wave		
$D = \frac{t_{on}}{T}$	$V_{avg} = V_p(D)$	$V_{rms} = V_p \sqrt{D}$
→ Triangular		
$i(t) = \frac{I_p(t)}{T}$	$I_{avg} = \frac{I_p}{2}$	$I_{rms} = \frac{I_p}{\sqrt{2}}$
→ Trapezoid		
$y = \frac{(I_p - I_{min})t + I_{min}}{DT} = D \left[\frac{I_p + I_{min}}{2} \right]$	$= \sqrt{\frac{D(I_p^2 + I_p I_{min} + I_{min}^2)}{3}}$	

Power formulae

→ Instantaneous Power	$p(t) = v(t) \times i(t)$	In sinusoidal
→ Avg Power	$P = \frac{1}{T} \int p(t) dt$	→ Effect of capacitor $i(t) = I_p \sin(\omega t + \pi/2)$
→ DC	$P = V_p I_p$	→ Effect of Inductor $i(t) = I_p \sin(\omega t - \pi/2)$
→ Rectangular	$P = V_p I_p \times D$	
→ Sinusoidal	$\frac{V_p I_p}{2}$	
→ RMR	$P = V_{rms} I_{rms}$	
→ $i(t) = \begin{cases} I_p \sin \theta \\ 0 \end{cases}$	$P = \frac{V_{supply} \times I_p}{\pi}$	

Low Power Audio Amplifier

$$f_1 = \frac{1}{2\pi R_L C} \text{ (cut-off frequency)}$$

$$\text{Gain} = A = \frac{2 \times 15k}{R_{pin} + 150\Omega} \text{ (1and8)}$$

Filter design :

$$C_{bass} > \frac{1}{2\pi f_1 (R/7)}$$

→ Cut-off frequency

$$\text{gain} = 2A_{mid}$$

$$A_{mid} = \frac{2 \times 15k \parallel 15k}{1.35 + 150}$$

→ for treble boost

$$R_{pin} \text{ (1and8)} = \frac{2 \times 15k \parallel 15k - 150}{20 \rightarrow \text{gain}}$$

→ for capacitor

$$C = \frac{1}{2\pi \times f_{treble} \times R_{pin} \text{ (1-8)}}$$

Operational Amp

Gain Bandwidth Product

$$GBW = A_o \times f_v$$

$$A_o = V_o / V_i$$

$$SR = \frac{dV_i}{dt}$$

$$\rightarrow \text{rectangular} : f_{\max} = \frac{SR}{2\pi V_p}$$

$$P_L = V_L \times I_L$$

$$P_{\text{supply}} = V_{\text{supply}} \times I_L(\text{DC})$$

$$P_{IC} = P_{\text{supply}} - P_L$$

$$\text{Max Power} = V_L = \frac{1}{2} V_{\text{supply}}$$

$$P_L = \frac{V_L^2}{R_L} ; P_L = I_L^2 \times R_L$$